

2022 H2 Prelim Paper 1 Solutions:

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
| B  | D  | A  | C  | D  | B  | C  | C  | B  | A  |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| D  | C  | A  | C  | D  | A  | C  | C  | A  | C  |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| B  | D  | B  | A  | D  | A  | D  | B  | C  | B  |

1 B

$$C_d = \frac{2F}{v^n \rho A} \Rightarrow C_d v^n \rho A = 2F$$

Unit of  $v = \text{m s}^{-1}$

Unit of  $\rho = \text{kg m}^{-3}$

Unit of  $A = \text{m}^2$

Unit of  $F = \text{kg m s}^{-2}$

$$\Rightarrow 1 \times \text{kg m}^{n-1} \text{s}^{-n} = \text{kg m s}^{-2}$$

$$\Rightarrow n = 2$$

2 D

0.12